

# Glasgow Coma Scale

## What is the Glasgow Coma Scale

Due to my unfamiliarity with the GCS vital sign, I decided that this would be a good time to explore what it means and its significance within the triage system. According to the Cleveland Clinic (2023), **GCS stands for the Glasgow Coma Scale**. It is a system used by healthcare providers to determine how conscious a person is, or how deeply they may be in a coma, based on eye, speech, and movement responses. The scale was created in 1974 by a group of experts at the University of Glasgow, which is where the name “Glasgow” originates from.

To calculate a patient’s Glasgow Coma Scale score, healthcare providers assess three categories and add the scores together. These categories are

### 1. Eye response: This relates to how awake and alert you are.

| Eye response score | Score meaning                                                                             |
|--------------------|-------------------------------------------------------------------------------------------|
| 4                  | You can open your eyes and keep them open on your own.                                    |
| 3                  | You only open your eyes when someone tells you to do so. Your eyes stay closed otherwise. |
| 2                  | Your eyes only open in response to feeling pressure.                                      |
| 1                  | Your eyes don’t open for any reason.                                                      |

Reprinted from *Glasgow Coma Scale (GCS): What it is, interpretation & chart*, by Cleveland Clinic, 2023, <https://my.clevelandclinic.org/health/diagnostics/24848-glasgow-coma-scale-gcs>

- 2. Motor response:** This part is about how well your brain can control muscle movement. It can also show if there are any issues with the connections between your brain and the rest of your body.

| Verbal response score | Score meaning                                                                                                 |
|-----------------------|---------------------------------------------------------------------------------------------------------------|
| 5                     | You're oriented. You can correctly answer questions about who you are, where you're at, the day or year, etc. |
| 4                     | You're confused. You can answer questions, but your answers show you're not fully aware of what's happening.  |
| 3                     | You can talk and others can understand words you say, but your responses to questions don't make sense.       |
| 2                     | You can't talk and can only make sounds or noises.                                                            |
| 1                     | You can't speak or make sounds.                                                                               |

Reprinted from *Glasgow Coma Scale (GCS): What it is, interpretation & chart*, by Cleveland Clinic, 2023, <https://my.clevelandclinic.org/health/diagnostics/24848-glasgow-coma-scale-gcs>

**3. Verbal response: This tests how well certain brain abilities work, including thinking, memory, attention span and awareness of your surroundings.**

| Motor (movement) response score | Score meaning                                                   |
|---------------------------------|-----------------------------------------------------------------|
| 6                               | You follow instructions on how and when to move.                |
| 5                               | You intentionally move away from something that presses on you. |
| 4                               | You only move away from something pressing on you as a reflex.  |
| 3                               | You flex muscles (pull inward) in response to pressure.         |
| 2                               | You extend muscles (stretch outward) in response to pressure.   |
| 1                               | You don't move in response to pressure.                         |

Reprinted from *Glasgow Coma Scale (GCS): What it is, interpretation & chart*, by Cleveland Clinic, 2023, <https://my.clevelandclinic.org/health/diagnostics/24848-glasgow-coma-scale-gcs>

## What is its Clinical Range?

Healthcare providers often repeat neurological examinations at regular intervals to monitor and document any changes in a patient's GCS score.

The **highest** possible GCS score is **15**, while the **lowest** is **3**. A score of 15 indicates that a patient is fully awake, responsive, and shows no issues with memory or thinking ability. Generally, a **score of 8 or below** suggests that the patient is in a coma, with lower scores representing deeper levels of unconsciousness (Cleveland Clinic, 2023).

## What does GCS mean for Triage?

The GCS is highly significant within healthcare because it reflects a patient's level of consciousness, which is essential in determining who requires urgent medical attention. A decreased level of consciousness is often a sign of a serious medical emergency. During triage, if a healthcare provider observes a GCS score lower than 8, the patient would typically be placed on high alert and require immediate medical attention. Furthermore, following Dr. Alemante's lecture yesterday, if two critically ill patients arrive at an under-resourced clinic, the patient with the lower GCS score may be prioritised over the patient with the higher score or may not be prioritized due to a low survival rate and attention is instead given to the other patient who may have a chance at survival.

However, it is important to note that the GCS is most significant in patients with neurological impairment. A patient may still have a high GCS score while also presenting with dangerously abnormal vital signs, such as a high MAP, which would also require urgent escalation to a physician. Therefore, while the GCS should not be used on its own to prioritise patients, it remains extremely valuable within the triage process, particularly from a neurological standpoint.

## Reference List:

Cleveland Clinic. (2023, March 26). *Glasgow Coma Scale (GCS): What it is, interpretation & chart*. Cleveland Clinic.

<https://my.clevelandclinic.org/health/diagnostics/24848-glasgow-coma-scale-gcs>

[OBJ]